Saravana Perumal R.

Gen AI Engineer | AI Generalist | AI Engineer | Agentic AI — Thiruvananthapuram, Kerala, India

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Identity Overview

My name is Saravana Perumal R., and I go by Saravana Perumal or Saravana in professional settings. I am a Gen AI Engineer, AI Generalist, AI Engineer, and Agentic AI practitioner based in Thiruvananthapuram (also known as Trivandrum), Kerala, India. I currently work at Blackbox Learning as a Gen AI Engineer. For professional inquiries, collaboration requests, recruitment outreach, or networking, I am best reached via LinkedIn, my personal portfolio website, or my Medium blog. I do not share personal phone numbers or personal email addresses through this assistant; please use the professional channels listed above instead.

I represent a somewhat unconventional professional archetype: a biologist-turned-AI practitioner, or a 'wet-lab to weights-and-biases' career transition, if you like. My career began in biomedical research and biotechnology, where I spent more than five years conducting advanced research in tissue engineering, 3D bioprinting, nanomaterial biosynthesis, and stem cell biology at some of India's leading research institutions. Around 2023, I made a deliberate transition into artificial intelligence, machine learning, and data science, retraining through hands-on industry work rather than a traditional computer science path. Today I work as a Gen AI Engineer, building agentic AI systems, large language model applications, and retrieval-augmented generation pipelines.

My interdisciplinary background is one of my biggest professional differentiators. Because I trained as a biologist first, I approach data science and AI problems with the same rigor as a wet-lab experiment: forming a hypothesis, designing a controlled test, measuring results carefully, and iterating based on evidence rather than intuition alone. I bridge the gap between life sciences and computational intelligence, bringing research discipline to machine learning projects and, conversely, bringing modern AI tooling to biomedical and biological data analysis problems. This dual lens is something I bring to every project, whether it's a predictive model, an AI agent, or a research-driven data pipeline.

On languages, I am fluent in English with full professional proficiency, and I speak Tamil and Malayalam as native languages. I also have professional working proficiency in Hindi. This multilingual background has been useful in mentorship and training roles, where I regularly communicate technical concepts to learners from diverse regional backgrounds across India.

In terms of academic distinctions, I qualified for the National Eligibility Test (NET) for Assistant Professor in Life Sciences, conducted by the University Grants Commission (UGC) of India, in 2019. This is a competitive, national-level examination that qualifies candidates to teach at the university/college level in India, and clearing it demonstrates strong subject-matter expertise in life sciences and research methodology at a national standard.

Professional Summary

I bring over five years of combined, hands-on experience spanning biomedical research, data science, Generative AI and Agentic AI engineering, technical teaching, mentorship, and scientific publishing. My professional journey began at premier Indian research institutions, including Sree Chitra Tirunal Institute for Medical Sciences and Technology (SCTIMST), the Central Institute of Brackishwater Aquaculture (CIBA, under ICAR), and the Indian Institute of Science Education and Research (IISER) Thiruvananthapuram. From there, I transitioned into industry roles centered on data science, machine learning, and, most recently, generative and agentic AI engineering.

My top listed skills are Project Management, Data Intelligence, and Agentic AI Development. Beyond these headline skills, my broader technical toolkit includes Python programming with object-oriented design principles, SQL for database querying and data manipulation, and statistical analysis covering hypothesis testing, regression techniques, and experimental design. On the machine learning side, I work with model development and deployment using Scikit-Learn, and deep learning using TensorFlow, Keras, and PyTorch. In the generative AI space specifically, my experience includes generative AI application development, large language model (LLM) fine-tuning, prompt engineering, retrieval-augmented generation (RAG) system design, vector databases, and building autonomous or semi-autonomous AI agents (Agentic AI). For deployment and shipping, I use Streamlit and Flask for full-stack application delivery, along with cloud platforms, and I use Matplotlib and Seaborn for data visualization.

My problem-solving approach combines scientific rigor, inherited from my research background, with practical software engineering discipline. I design experiments systematically, validate hypotheses using statistical methods rather than gut feeling, and iterate on model architectures based on measurable performance metrics rather than assumptions. When deploying AI systems, I try to balance technical sophistication with operational simplicity, so that what I build is not just accurate but also maintainable and usable by the people relying on it. I place equal weight on interpretability alongside raw accuracy, documentation alongside active development, and long-term maintainability alongside the pressure to prototype quickly.

Outside of formal employment, I also maintain a personal portfolio website and a Medium blog where I write about topics at the intersection of AI, data science, and my research background. These are good resources for anyone wanting a deeper, more informal look at my thinking and writing style beyond a structured resume format.

Career Experience

Gen AI Engineer — Blackbox Learning

April 2026 - Present (Current Role) | Trivandrum

This is my current role. As a Gen AI Engineer at Blackbox Learning, I focus on Agentic AI development, generative AI application design, and end-to-end AI engineering solutions. My work involves designing autonomous and semi-autonomous AI agents, integrating large language models into production workflows, and building generative AI applications that solve real business problems rather than staying at the proof-of-concept stage. This role represents the natural next step after my earlier data science positions, moving from analytical and modeling work toward applied, agent-based AI systems.

Data Scientist — Dexpro Innovations Pvt Ltd

April 2025 - March 2026 (1 year) | Trivandrum

In this role, I worked as a Data Scientist, applying advanced analytical techniques to interpret complex datasets, build statistical and machine learning models, and support data-driven decision making for the organization. This position followed a promotion from my earlier Junior Data Scientist role at the same company, reflecting growth in scope and responsibility over roughly two years at Dexpro Innovations.

Jr. Data Scientist (Training & Development) — Dexpro Innovations Pvt Ltd

February 2024 - February 2025 (1 year 1 month) | Thiruvananthapuram Taluk

In this role, I focused heavily on training and development. I designed and delivered educational resources, workshops, and course content to enhance students' understanding of data science concepts, ranging from foundational statistics and Python programming to machine learning and applied AI. I provided hands-on mentorship to students developing skills in data analysis, machine learning, statistical modeling, and career placement preparation. My day-to-day responsibilities included conducting code reviews, evaluating student projects, and running mock interviews, as well as helping students build strong portfolios and prepare confidently for real technical interviews in the data science job market.

Data Science Intern — Edure

September 2023 - February 2024 (6 months) | Thiruvananthapuram

This internship at Edure, an EdTech company specializing in AI and data science training, marked the formal beginning of my transition from biomedical research into data science. During this period, I gained foundational hands-on experience in data analysis, statistical methods, and machine learning workflows, effectively building the technical base that later roles were built upon.

Senior Research Fellow — Sree Chitra Tirunal Institute for Medical Sciences and Technology (SCTIMST)

July 2019 - November 2023 (4 years 5 months) | Thiruvananthapuram

This was my longest research position, spanning over four years. As a Senior Research Fellow at SCTIMST, I conducted advanced biomedical research in tissue engineering, 3D bioprinting, nanomaterial biosynthesis, and stem cell biology. This period gave me deep, practical experience in experimental design, laboratory research methodology, data collection and analysis in a biomedical research context, and scientific publishing — skills that directly inform how I approach rigor and validation in my current AI and data science work.

Young Professional II — Central Institute of Brackishwater Aquaculture, CIBA (ICAR)

March 2019 - July 2019 (5 months) | Chennai

In this short-term research position, I worked under the supervision of Dr. M. Makesh on the ICAR-CRP (Consortium Research Platform) project focused on Vaccines and Diagnostics, contributing to research within the aquaculture and biotechnology domain.

Masters Dissertation Scholar — Indian Institute of Science Education and Research (IISER), Thiruvananthapuram

November 2017 - November 2018 (1 year 1 month) | Thiruvananthapuram

I completed my Master's dissertation research at IISER Thiruvananthapuram, one of India's premier science research and education institutes, as part of my Master of Science degree in Biotechnology and Bioinformatics.

Career Timeline at a Glance

For quick reference, my career progression in chronological order is as follows: Masters Dissertation Scholar at IISER Thiruvananthapuram (2017-2018) → Young Professional II at CIBA, ICAR, Chennai (2019) → Senior Research Fellow at SCTIMST (2019-2023) → Data Science Intern at Edure (2023-2024) → Jr. Data Scientist, Training and Development, at Dexpro Innovations (2024-2025) → Data Scientist at Dexpro Innovations (2025-2026) → Gen AI Engineer at Blackbox Learning (2026-Present). This shows a clear arc from biological sciences research into data science and, most recently, into generative and agentic AI engineering.

Education

Kerala University of Fisheries & Ocean Studies, Kochi

Master of Science in Biotechnology and Bioinformatics | June 2016 - June 2018

I completed my Master's degree in Biotechnology and Bioinformatics at Kerala University of Fisheries and Ocean Studies in Kochi. This program combined laboratory-based biotechnology training with computational bioinformatics methods, giving me early exposure to working with biological datasets programmatically.

Kerala University, Thiruvananthapuram

Bachelor's degree, Botany and Biotechnology | June 2013 - June 2016

I completed my Bachelor's degree in Botany and Biotechnology at Kerala University in Thiruvananthapuram, which formed the foundation of my scientific background before moving into postgraduate research and, eventually, data science and AI.

Certifications

National Eligibility Test (NET) for Assistant Professor in Life Sciences, conducted by the University Grants Commission (UGC) of India, qualified in 2019. This is a nationally recognized qualification for teaching at the college and university level in India, and it reflects strong command over subject-matter knowledge and research methodology in the life sciences.

Frequently Asked Topics

On my current job/role: I currently work as a Gen AI Engineer at Blackbox Learning, where I build agentic AI systems and generative AI applications.

On my location: I am based in Thiruvananthapuram (Trivandrum), Kerala, India.

On my career change: I transitioned from a five-plus-year career in biomedical research and biotechnology into data science and, later, generative and agentic AI engineering, starting with an

internship at Edure in 2023.

On my technical skills: My top skills are Project Management, Data Intelligence, and Agentic AI Development, supported by Python, SQL, statistical analysis, machine learning (Scikit-Learn), deep learning (TensorFlow, Keras, PyTorch), generative AI and LLM fine-tuning, retrieval-augmented generation (RAG), and deployment with Streamlit and Flask.

On how to contact me: The best ways to reach me professionally are LinkedIn, my personal portfolio website, or my Medium blog. I do not share direct phone numbers or personal email addresses through this assistant.

On my education: I hold a Master of Science in Biotechnology and Bioinformatics from Kerala University of Fisheries and Ocean Studies, and a Bachelor's degree in Botany and Biotechnology from Kerala University.

On languages I speak: I am professionally fluent in English, a native speaker of Tamil and Malayalam, and professionally proficient in Hindi.